

Bulk Cobordism Synthesis by Entanglement-Sourced Regge Sampling

Theory and Implementation Plan

Andrew Kelleher

Abstract

We develop the theory of, and an implementation plan for, a single construction: given a quantum operation $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$ between two boundary systems, *synthesize a bulk geometry* that realizes it. The bulk is a simplicial complex whose metric is the mutual information between quantum subsystems, whose dynamics is the discrete gravitational (Regge) action, and whose boundary is held *fixed* while the interior is rewritten by Monte Carlo moves. Realizability is the statement that a prescribed boundary state can be made a harmonic mode of the bulk Laplacian; we minimize the corresponding residual while holding the boundary fixed. The document is written as a sequence of self-contained tutorials on each mathematical object (mutual information and the information metric, simplicial complexes, the Choi–Jamiołkowski isomorphism, the Hodge Laplacian, the Regge action, Pachner moves, the heat-kernel spectral dimension), followed by the synthesis theory (the gravitational posterior, feasibility witnesses, and the free-energy certificate of obstruction) and a concrete implementation plan against the existing codebase. The emphasis throughout is theoretical. Honest scope statements, prior-art positioning (the mutual-information metric is due to Cao–Carroll–Michalakis), and the evidentiary bar for any emergent-dimension claim are stated explicitly.

Contents

1	Overview	1
1.1	The problem	1
1.2	The construction in one equation	2
1.3	Map of the document	2
2	Mathematical preliminaries: a tutorial on each element	2
2.1	Quantum states, reduced density matrices, partial trace	2
2.2	Entanglement entropy and mutual information	3
2.3	The mutual-information metric	3
2.4	Simplicial complexes: the discrete bulk	4
2.5	The Choi–Jamiołkowski isomorphism: bending an operator to a state	5
2.6	The Hodge Laplacian and its harmonics	5
2.7	The realizability residual is a quantum variance	6
2.8	Regge calculus: curvature without coordinates	6
2.9	Pachner moves and boundary-fixed ergodicity	7
2.10	The heat-kernel spectral dimension	8

3	The synthesis theory	9
3.1	The fixed-boundary problem	9
3.2	The gravitational posterior	9
3.3	Realizability: certification by a witness	10
3.4	Obstruction: certification by free energy	10
3.5	Summary of the theory	11
4	Implementation plan	11
4.1	Existing components	12
4.2	Workstream: boundary pinning in the sampler	12
4.3	Workstream: the residual term in acceptance	12
4.4	Workstream: the move set	12
4.5	Workstream: diagnostics and certificates	13
4.6	Experimental protocol	13
5	Honest assessment: novelty and the dimension bar	13

1 Overview

1.1 The problem

A topological quantum field theory assigns, to each boundary surface Σ , a Hilbert space $Z(\Sigma)$, and to each bulk W with $\partial W = \Sigma$ a vector $Z(W) \in Z(\Sigma)$. A morphism between two boundaries A, B is, in this language, a cobordism: a bulk W_{AB} interpolating between them. The inverse question is the subject of this document:

Given an operation $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$ relating two boundary systems, construct a bulk W_{AB} that realizes it — rather than assuming one is given.

Our strategy is to make “realizes” precise and computable. We bend U into a boundary *state* ψ_U (Choi–Jamiołkowski), build a discrete bulk whose geometry is sourced by quantum entanglement, and demand that ψ_U be a harmonic (zero) mode of the bulk’s Hodge Laplacian on the prescribed, fixed boundary. The degree to which this can be achieved is measured by a residual r ; the central computational task is

$$\boxed{\text{minimize the residual } r \text{ while holding the boundary fixed.}} \quad (1)$$

1.2 The construction in one equation

All of the theory below is organized around a single probability measure on bulk geometries K whose boundary subcomplex $L = \partial K$ is held fixed:

$$\pi_{\tau, \beta}(K) \propto \exp \left[-\beta S_{\text{Regge}}(K) - \frac{1}{\tau} r_U(K) \right], \quad K|_{\partial} = L. \quad (2)$$

Here S_{Regge} is the discrete Einstein–Hilbert action (a *gravitational prior* on geometries), $r_U(K)$ is the realizability residual (a *likelihood* that the geometry carries the operation U), β is the gravitational inverse temperature, and $\tau \downarrow 0$ sharpens the measure onto realizing geometries. Sampling (2) by Markov chain Monte Carlo, with moves that never touch the fixed boundary, is the synthesis algorithm. Each symbol is defined and motivated in its own tutorial section below; the synthesis theory (Section 3) explains why this particular measure is the right object and what its samples certify.

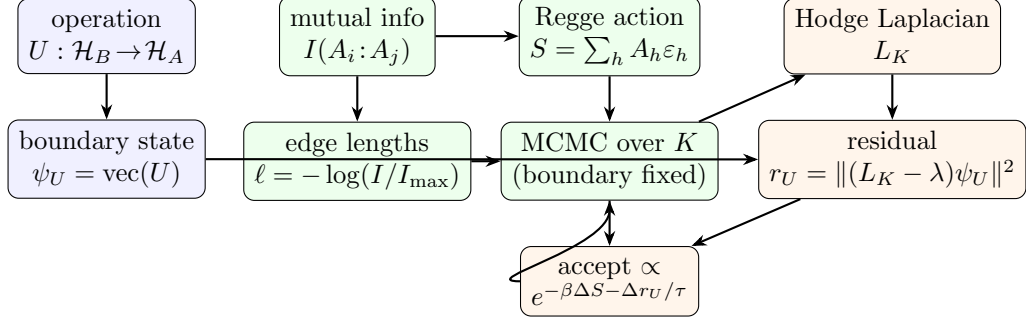


Figure 1: The synthesis pipeline. The operation is bent to a boundary state (left). The quantum subsystems’ mutual information sets the simplicial edge lengths and the Regge action (centre). Monte Carlo rewrites the interior at fixed boundary; each geometry yields a Hodge Laplacian whose harmonic is compared to ψ_U , and the residual feeds back into the acceptance (right).

1.3 Map of the document

Section 2 is the tutorial core: a self-contained pass over every mathematical element. Section 3 assembles them into the synthesis theory. Section 4 is the implementation plan against the existing code. Section 5 states the honest novelty and 4D-evidence positions. A reader who only wants the theory can stop after Section 3.

2 Mathematical preliminaries: a tutorial on each element

This section introduces, from first principles, each object appearing in (2) and Figure 1. Each subsection states a definition, gives the intuition, names the role the object plays in the construction, and flags the one caveat that matters.

2.1 Quantum states, reduced density matrices, partial trace

A quantum system is a complex Hilbert space $\mathcal{H}$ with a chosen orthonormal basis $\{|i\rangle\}$. A pure state is a unit vector $|\psi\rangle \in \mathcal{H}$; a general (possibly mixed) state is a *density matrix* ρ : a positive semidefinite operator with $\text{Tr } \rho = 1$. A pure state corresponds to the rank-one projector $\rho = |\psi\rangle\langle\psi|$.

For a bipartite system $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, the state of subsystem A alone is the *reduced density matrix*, obtained by the *partial trace* over B :

$$\rho_A = \text{Tr}_B \rho_{AB} = \sum_b (\mathbb{I}_A \otimes \langle b|) \rho_{AB} (\mathbb{I}_A \otimes |b\rangle). \quad (3)$$

The partial trace is the unique operation consistent with the requirement that local measurements on A are reproduced by ρ_A .

Role. In the construction, the “quantum subsystems” are factors of a large state produced by an interaction history; their reduced density matrices are the raw material from which entanglement — and hence geometry — is computed (Section 2.2). In the codebase these are the per-vertex ρ and per-pair joint ρ_{AB} tracked during the Monte Carlo.

2.2 Entanglement entropy and mutual information

Definition 2.1 (von Neumann entropy). The entropy of a state ρ is $S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_k p_k \log p_k$, where $\{p_k\}$ are the eigenvalues of ρ .

For a *pure* global state, $S(\rho_A) = S(\rho_B)$ quantifies the entanglement between A and its complement; it vanishes iff the state factorizes ($|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$).

Definition 2.2 (mutual information). For subsystems A, B the mutual information is

$$I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0, \quad (4)$$

with equality iff $\rho_{AB} = \rho_A \otimes \rho_B$ (no correlation, classical or quantum).

Mutual information is the total correlation between two subsystems. It is symmetric, nonnegative, and — crucially for what follows — it is large when subsystems are strongly correlated and small when they are nearly independent. It upper-bounds all two-point correlators (a consequence of subadditivity), so “high mutual information” literally means “these two subsystems strongly influence one another.”

Role. Mutual information is the bridge from quantum dynamics to geometry: two subsystems that are strongly correlated will be *close* in the emergent metric (Section 2.3). This is the Van Raamsdonk slogan “entanglement is the glue of spacetime” [1] made quantitative.

2.3 The mutual-information metric

We turn correlation into distance. The defining choice is a monotone decreasing map from mutual information to an edge length: strong correlation $\mapsto$ short edge.

Definition 2.3 (information length). For an edge $e = (i, j)$ between subsystems carrying mutual information I_{ij} ,

$$\ell_e = -\log\left(\frac{I_{ij}}{I_{\max}}\right), \quad (5)$$

with $I_{\max}$ a normalization (the maximal attainable mutual information) and a floor $\ell_e \leq -\log \varepsilon$ enforced when $I_{ij} < \varepsilon I_{\max}$.

As $I_{ij} \rightarrow I_{\max}$ the length $\ell_e \rightarrow 0$ (maximally correlated subsystems are coincident); as $I_{ij} \rightarrow 0$ the length diverges (uncorrelated subsystems are infinitely far apart). The squared lengths ℓ_e^2 are the dynamical variables of the discrete geometry.

Remark 2.4 (prior art — positioning). The functional form (5) is *not original to this work*. It is exactly the ansatz of Cao, Carroll, and Michalakis [2], who take an edge weight $\propto \Phi(I/I_0)$ with $\Phi(x) = -\log x$ on normalized mutual information, and reconstruct geometry from it by multidimensional scaling. Any presentation of the present construction must cite [2] for the metric and locate its own contribution elsewhere (Section 5): in the *sampled* Regge dynamics, the *fixed-boundary* synthesis, and the *inverse* (operation-realizing) direction, none of which appear in [2].

Caveat. Mutual-information distance is not automatically a metric. The triangle inequality can fail; on 1D quantum chains it holds only within a window of the correlation-decay exponent [3], and fails in gapped/exponentially-correlated phases. Because the Regge action below requires non-degenerate simplices (valid triangle inequalities on each cell), the realized edge lengths (5) must be checked for simplicial validity; the floor $-\log \varepsilon$ mitigates but does not guarantee it.

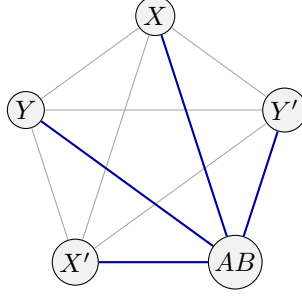


Figure 2: The interaction “bowtie”: two inputs X, Y produce two local marginals X', Y' and a joint hub AB , gluing in a 4-simplex (K_5 , ten edges). Hub spokes (blue) carry the joint mutual information; worldline self-edges carry the input entropies; the three cross-worldline edges are set to zero mutual information in the local-frame model.

2.4 Simplicial complexes: the discrete bulk

Definition 2.5 (abstract simplicial complex). A simplicial complex K on a vertex set V is a collection of finite subsets (simplices) of V , closed under taking subsets. A k -simplex has $k + 1$ vertices; 0-simplices are vertices, 1-simplices edges, 2-simplices triangles, 3-simplices tetrahedra, and a 4-simplex has 5 vertices and $\binom{5}{2} = 10$ edges.

A *geometric* (Regge) simplicial complex additionally assigns a squared length ℓ_e^2 to each edge; this is the discrete metric. The codimension-one faces lying in exactly one top-dimensional simplex form the *boundary* ∂K ; the rest is the *interior*. This boundary/interior partition is the structural object we will pin in (1).

Example 2.6 (the interaction bowtie is a 4-simplex). In the interaction-history model, a pairwise interaction of systems X, Y produces three outputs — the two local marginals X', Y' and a joint hub AB — and glues in the 5-vertex cell $\{X, Y, X', AB, Y'\}$ with all 10 edges. This is precisely (the 1-skeleton of) a 4-simplex; each interaction contributes one unit of local 4-volume. See Figure 2.

2.5 The Choi–Jamiołkowski isomorphism: bending an operator to a state

To ask whether a *geometry* carries an *operation*, we first convert the operation into a state living on the boundary.

Definition 2.7 (vectorization / Choi–Jamiołkowski). For a linear operator $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$ with matrix entries U_{ij} in fixed bases, its vectorization is

$$\psi_U = \text{vec}(U) = \sum_{i,j} U_{ij} |i\rangle_A \otimes |j\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B. \quad (6)$$

In row-major storage the flat index $i d_B + j$ of the tensor coincides with the flat index of U , so vectorization is a reinterpretation of the same array of numbers.

The map $U \mapsto \psi_U$ is a linear isomorphism $\text{Hom}(\mathcal{H}_B, \mathcal{H}_A) \cong \mathcal{H}_A \otimes \mathcal{H}_B$. It carries operator composition into state overlaps; in particular the transition amplitude through U is a boundary inner product,

$$\langle \psi_A | U | \psi_B \rangle = \langle \text{vec}(|\psi_A\rangle\langle\psi_B|), \psi_U \rangle. \quad (7)$$

This identity is what lets a single boundary *vector* ψ_U stand in for the entire operation when we test realizability, and it is the bridge to the partition-function reading $Z(W) = \langle \psi_A | U | \psi_B \rangle$.

Role. ψ_U is the target the bulk must carry: the prescribed boundary data in (1). It is pinned on the boundary support throughout the synthesis.

2.6 The Hodge Laplacian and its harmonics

The operator the bulk geometry induces, and whose modes we match to ψ_U , is the Hodge Laplacian. We need it at degree $k = 0$ (functions on vertices) and $k \geq 1$ (cochains on edges, triangles, ...).

Degree zero (the weighted graph Laplacian). On the 1-skeleton, with a Hermitian weighted adjacency $A_{ij} = \sum_{(i,j)} w_{ij} e^{i\theta_{ij}}$ (weight $w_{ij} = \ell_{ij}^2$, an optional $U(1)$ phase θ_{ij} carried with opposite sign on the reverse orientation so $A = A^\dagger$) and degree $D_{ii} = \sum_j |w_{ij}|$,

$$L = D - A, \quad L = L^\dagger, \quad L \succeq 0. \quad (8)$$

Hermiticity makes e^{-iLt} unitary (a discrete time evolution) and the spectrum real.

Higher degree (the metric Hodge Laplacian). Let ∂_k be the integer boundary maps of the chain complex and $W_k = \text{diag}(k\text{-cell volumes})$ the diagonal inner-product (mass) weights, with $W_0 = \mathbb{K}$. The metric adjoint is $\partial_k^* = W_k^{-1} \partial_k^\top W_{k-1}$ and

$$L_k = \partial_k^* \partial_k + \partial_{k+1} \partial_{k+1}^*. \quad (9)$$

Theorem 2.8 (discrete Hodge decomposition; Eckmann). $\ker L_k \cong H_k(K)$, the k -th homology. Hence $\dim \ker L_k = b_k$, the k -th Betti number, for any positive weights. The cochain space splits orthogonally as $\text{im } \partial_{k+1} \oplus \ker L_k \oplus \text{im } \partial_k^*$.

The elements of $\ker L_k$ are the *harmonic* cochains. Theorem 2.8 has a sharp consequence for us: the *number* of harmonics is a topological invariant, but *which* cochains are harmonic depends on the metric (the weights W_k), i.e. on the edge lengths (5). Synthesis tunes the metric so that the prescribed ψ_U becomes harmonic.

Degree controls which simplices are visible. With unit weights ($W_k = \mathbb{K}$) the construction reduces to the *combinatorial* Hodge Laplacian $L_k = \partial_k^\top \partial_k + \partial_{k+1} \partial_{k+1}^\top$; at degree one, $L_1 = \partial_1^\top \partial_1 + \partial_2 \partial_2^\top$. A degree-dependence that becomes decisive in Sections 2.9 and 4: the degree-zero operator $L_0 = D - A$ reads *only the 1-skeleton*, so 2- and higher-dimensional simplices are spectrally invisible at $k = 0$; whereas L_1 depends on the 2-simplices through ∂_2 — a *filled* triangle and a *hollow* one give the same L_0 but different L_1 . Thus the “Hodge-qubit” degree $k = 1$ (the harmonic 1-forms, $\dim \ker L_1 = b_1$) is the relevant degree for the cobordism/Dijkgraaf–Witten bridge, and synthesizing at $k = 1$ requires growth that can add genuinely 2-dimensional content, not merely edges.

Caveat. At $k \geq 1$ the adjoint (9) contains W_k^{-1} — an inverse mass matrix. Since the W_k are simplex volumes (square roots of Cayley–Menger polynomials in ℓ^2), L_k depends *rationally* on the metric. This non-affine dependence is the analytic obstruction to convex formulations of the matching problem and is one motivation for the sampling approach of Section 3.

2.7 The realizability residual is a quantum variance

We now define “ ψ_U is (approximately) a harmonic/eigenmode of L .” Let $P_\psi = \psi \psi^\dagger / \|\psi\|^2$ be the orthogonal projector onto ψ .

Definition 2.9 (residual).

$$r(\psi) = \|(\mathbb{I} - P_\psi) L\psi\|^2. \quad (10)$$

Geometrically, r measures the part of $L\psi$ not parallel to ψ : it is zero exactly when $L\psi = \lambda\psi$ for some λ , i.e. when ψ is an eigenvector. A short computation gives the identity that organizes the entire optimization analysis:

Proposition 2.10. *With $\lambda = \langle \psi | L | \psi \rangle / \|\psi\|^2$ the Rayleigh quotient,*

$$r(\psi) = \|L\psi - \lambda\psi\|^2 = \|\psi\|^2 \left(\langle L^2 \rangle_\psi - \langle L \rangle_\psi^2 \right) = \|\psi\|^2 \text{Var}_\psi(L). \quad (11)$$

That is, **the residual is the quantum variance of the Hermitian operator L in the state ψ** . It vanishes iff ψ has definite “energy” under L — iff it is an eigenvector. For the harmonic target ($\lambda = 0$) it reduces to $r = \|L\psi\|^2$.

Remark 2.11 (why the optimization is hard, in one line). As a function of the state ψ on the unit sphere, $\text{Var}_\psi(L)$ is a Morse–Bott function whose critical points are *all* eigenvectors of L ; its near-zero sublevel sets are disconnected (one component per distinct eigenvalue). As a function of the edge weights w (with ψ, θ fixed) it is convex — the Hessian is a Gram/covariance matrix — but jointly in (w, θ, ψ) the term $L\psi$ is bilinear and the phases enter through $e^{i\theta}$, so the landscape is globally non-convex. This is why we do not minimize r by gradient descent over weights but *sample* (2) over geometries (Section 3).

2.8 Regge calculus: curvature without coordinates

Regge calculus is general relativity on a simplicial complex, with the squared edge lengths as the gravitational field [4]. Curvature is concentrated on the codimension-two faces, called *hinges* (edges in 3D, triangles in 4D).

Definition 2.12 (deficit angle). Around a hinge h , the dihedral angles $\theta_h^{(\sigma)}$ of the top simplices σ meeting h do not in general sum to the flat value 2π . The shortfall

$$\varepsilon_h = 2\pi - \sum_{\sigma \supset h} \theta_h^{(\sigma)} \quad (12)$$

is the deficit angle; it is the discrete curvature carried by h . Dihedral angles are computed from the Gram matrix of each simplex (itself affine in the squared edge lengths) via its cofactors.

Definition 2.13 (Regge action). The discrete Einstein–Hilbert action is

$$S_{\text{Regge}} = \sum_h A_h \varepsilon_h \quad (+ \text{matter, cosmological, and boundary terms}), \quad (13)$$

where A_h is the volume of the hinge. Its stationary points (with respect to the edge lengths) are the discrete Einstein equations. A point mass in 3D appears as a conical deficit, $\varepsilon = 8\pi Gm$: matter is curvature.

Role. S_{Regge} is the gravitational prior in (2): it weights geometries by their discrete Einstein–Hilbert action, so the synthesized bulk is not an arbitrary graph that fits ψ_U but one that also tends to extremize the gravitational action. This is what makes the construction a discrete-gravity statement rather than a curve fit.

Caveat. The Euclidean gravitational action is unbounded below (the conformal-mode problem [6]); minimizing S_{Regge} directly diverges. In the sampling frame this is handled by working at *fixed volume* (a cap on the number of cells) and with a cosmological term, exactly as in dynamical triangulations; one samples $e^{-\beta S}$ rather than minimizing S .

A boundary term for a fixed boundary. When the boundary is held fixed (Dirichlet data), the variational problem is well-posed only with the Gibbons–Hawking–York boundary term $S_{\text{GHY}} \propto \int_{\partial} \sqrt{h} K$ (extrinsic curvature) [5]; its Regge analogue is a sum of boundary hinge terms. This is the discrete shadow of the Hartle–Hawking prescription “fix the boundary, sum over interior geometries” [7].

2.9 Pachner moves and boundary-fixed ergodicity

To sample geometries we need moves that rewrite the triangulation. The canonical local moves are the Pachner (bistellar) moves.

Definition 2.14 (Pachner moves). In dimension d , the (p, q) bistellar move replaces a configuration of p top simplices sharing a sub-simplex by the complementary configuration of q , with $p + q = d + 2$ (e.g. in 3D: $1 \leftrightarrow 4$, $2 \leftrightarrow 3$). Each move preserves the piecewise-linear homeomorphism type.

Theorem 2.15 (Pachner; Casali for boundaries). *Any two triangulations of the same PL manifold are connected by a finite sequence of bistellar moves [9]. Moreover, if two triangulations coincide on the boundary, they are connected by bistellar moves supported in the interior alone [10].*

Theorem 2.15 is the mathematical license for (1): restricting the Monte Carlo to interior-only Pachner moves keeps the chain *ergodic* over all admissible fillings while *never touching the boundary*. The fixed boundary is thus encoded in the move set, not enforced by rejection.

Which boundary-fixed moves are spectrally active — a degree-dependence. Not every boundary-fixed growth move changes L_k , and at the Hodge-qubit degree this is sharp. With a bit-exact pinned boundary on a pure d -complex, the additive (free) attach of a single interior vertex falls into one of three cases (internal work, #203):

- (i) a *dangling* additive edge or triangle is dropped by the chain complex’s downward-closure construction, leaving L_k — and hence the residual — bit-exact: spectrally *inert*;
- (ii) an additive *top-cell* attach is *boundary-locked* — it introduces boundary edges incident to the new vertex, which the bit-exact ∂W guard rejects;
- (iii) the stellar Pachner subdivision (coning a vertex into a top cell) is the *one* boundary-fixed move that enriches L_k .

The practical consequence is a reversal between degrees. Letting interior connectivity be a free variable (the free-connectivity engine, #201) strictly *enlarges* the realizable set at $k = 0$ — the “fan” win, where a single non-cone interior vertex couples boundary leaves that no cone can reach. But at $k \geq 1$ it *collapses*: the extra edge/triangle candidates are inert or locked, so **free connectivity equals the cone**. The 2-cell is genuinely spectrally real at $k = 1$ (it shifts the $k = 1$ residual while leaving $k = 0$ bit-exact); the additive engine simply cannot realize it under the pinned boundary. Hence at the cobordism/Hodge-qubit degree the spectrally-active boundary-fixed move is the Pachner cone, and the move set must reflect this (Section 4).

Caveat. Casali’s ergodicity (Theorem 2.15) is stated for PL *manifolds*. The free-connectivity engine deliberately drops the manifold constraint: `attachInteriorVertex` validates only that the complex stays a downward-closed abstract simplicial complex and that ∂W stays bit-exact — no manifold, pseudomanifold, or purity condition — so topology is emergent (#201). On such general (non-manifold) complexes the manifold-based connectivity theorem does not directly apply; an ergodicity guarantee for the relaxed move set is an open item, distinct from the manifold case below.

Caveat. Casali’s theorem is topological (PL category). The *geometric* (constant-curvature) version of interior-fixed connectivity can require subdivisions in general, and is established directly only in low dimensions [11]. Match the move set to the category the bulk lives in.

2.10 The heat-kernel spectral dimension

Finally, the observable used to characterize the emergent geometry. Run a diffusion on the complex with the graph Laplacian as generator; the return probability is

$$P(\sigma) = \frac{1}{|V|} \text{Tr } e^{-\sigma L}, \quad d_S(\sigma) = -2 \frac{d \log P(\sigma)}{d \log \sigma}. \quad (14)$$

Definition 2.16 (spectral dimension). $d_S(\sigma)$ is the spectral dimension at diffusion scale σ . On a smooth d -manifold $d_S \equiv d$; on fractal or quantum geometries it runs with scale.

The trace is evaluated by a Krylov–Lanczos heat-kernel method, d_S by a Savitzky–Golay local fit, and an Ambjørn–Loll form $d_S(\sigma) = d_\infty - C/(B + \sigma)$ summarizes the running.

Role. The spectral dimension is the *diagnostic* for whether the synthesized ensemble looks like a (3+1)-dimensional spacetime at large scales. It is a secondary observable, not the optimization target.

Caveat. The evidentiary bar for “emergent 4D” is high (Section 5): a single peak near $d_S = 4$ is not sufficient. The field requires a *plateau*, the *ultraviolet running toward* $d_S \approx 2$ seen across approaches [19, 20], a corroborating Hausdorff dimension, and finite-size scaling.

3 The synthesis theory

We now assemble the elements into the theory behind (2): what the fixed boundary means, why the posterior is the right object, and — the technical heart — what its samples certify, both for realizability and for its negation.

3.1 The fixed-boundary problem

The conceptual template is the gravitational path integral with Dirichlet data: fix the boundary geometry/state and integrate over bulks that fill it. Three classical statements say the same thing at increasing levels of precision:

- (i) *Gibbons–Hawking–York* [5]: the boundary term that makes “fixed boundary metric” a well-posed variational condition.
- (ii) *Hartle–Hawking* [7]: the wavefunction $\Psi[h, \Sigma] = \int_{g|_{\Sigma=h}} \mathcal{D}g e^{-S_E[g]}$ — fix Σ , sum over interiors.
- (iii) *Atiyah* [8]: the functorial form — a boundary Σ has a state space $Z(\Sigma)$, and a bulk W with $\partial W = \Sigma$ is a vector $Z(W) \in Z(\Sigma)$.

Our discrete realization fixes a boundary subcomplex L carrying the prescribed state ψ_U , and rewrites the interior by the boundary-preserving moves of Theorem 2.15.

The correspondence hypotheses being tested. Synthesis is the engine for three *State–Operation–Cobordism* hypotheses (internal issue #202), stated for an operation U between prepared boundary states (ψ_A, ψ_B) :

(H1) **Realizable**

$W_{AB} = \text{geo}(U)$ exists — U is realizable, $r_U \rightarrow 0$ under boundary-fixed growth.

(H2) **Boundary**

$\partial W_{AB} = \overline{\text{geo}(\psi_B)} \sqcup \text{geo}(\psi_A)$ — the boundary is the two states, held bit-exact by construction.

(H3) **Amplitude**

$Z(W_{AB}) = \langle \psi_A | U | \psi_B \rangle$ — the realized geometry reproduces the transition amplitude, checked *explicitly* rather than assumed from $r_U \rightarrow 0$.

The live empirical question (#202) is whether the spectral/free-connectivity realizable set strictly *exceeds the discrete Dijkgraaf–Witten S_3 shadow* — realizing operations the topological state-sum cannot — and whether H3 nonetheless holds on that excess, which would exhibit the spectral layer as a genuine continuum extension of the topological one. Per Section 2.9 this test is degree-sensitive: at $k = 0$ free connectivity can enlarge the realizable set, while at the Hodge-qubit degree $k = 1$ it reduces to the Pachner cone.

3.2 The gravitational posterior

Definition 3.1 (synthesis measure). On the space $\mathcal{K}_L$ of bulk geometries with boundary L fixed,

$$\pi_{\tau, \beta}(K) = \frac{1}{\mathcal{Z}_{\tau, \beta}} \exp \left[-\beta S_{\text{Regge}}(K) - \frac{1}{\tau} r_U(K) \right], \quad \mathcal{Z}_{\tau, \beta} = \sum_{K \in \mathcal{K}_L} e^{-\beta S_{\text{Regge}}(K) - r_U(K)/\tau}. \quad (15)$$

Read as Bayesian inference, $e^{-\beta S_{\text{Regge}}}$ is a *prior* over geometries (discrete gravity) and $e^{-r_U/\tau}$ is the *likelihood* that a geometry realizes U ; $\pi_{\tau, \beta}$ is the posterior, and the maximum-a-posteriori geometry is the synthesized bulk. The annealing parameter $\tau \downarrow 0$ concentrates the posterior on minimizers of r_U (Theorem 3.5).

Sampling. We draw from $\pi_{\tau, \beta}$ by Metropolis–Hastings with interior-only Pachner moves and interior edge-length updates. The acceptance probability for $K \rightarrow K'$ is

$$\alpha(K \rightarrow K') = \min \left\{ 1, \frac{q(K' \rightarrow K)}{q(K \rightarrow K')} e^{-\beta \Delta S_{\text{Regge}} - \Delta r_U / \tau} \right\}, \quad (16)$$

where the proposal ratio $q(K' \rightarrow K)/q(K \rightarrow K')$ (the Hastings correction) is essential because the boundary-preserving moves have asymmetric, configuration-dependent proposal probabilities. Detailed balance with respect to $\pi_{\tau, \beta}$ then holds, and by Theorem 2.15 the chain is irreducible on $\mathcal{K}_L$.

3.3 Realizability: certification by a witness

Proposition 3.2 (feasibility witness). *If the sampler produces any $K^* \in \mathcal{K}_L$ with $r_U(K^*) \leq \epsilon$, then K^* is a constructive certificate that ψ_U is realizable to tolerance ϵ : the bulk K^* carries ψ_U as an ϵ -approximate harmonic of L_{K^*} with the boundary byte-identical to L .*

This is the easy direction: existence is proved by exhibition. Two structural facts qualify it.

Remark 3.3 (non-uniqueness). A witness certifies *that* a filling exists, not *which*. Inverse spectral geometry is emphatic that boundary/spectral data generally does not determine the interior — one “cannot hear the shape of a drum” [12]. Uniqueness of reconstruction requires strong structural hypotheses; for the graph Laplacian, the Gel’fand inverse problem is solvable when each boundary vertex attaches to exactly one interior vertex (plus an extreme-point condition) [13].

Remark 3.4 (local realizability is linear). For a *fixed* combinatorial graph and a fixed harmonic target ($\lambda = 0$), the condition $L\psi_U = 0$ is linear in the edge weights; existence of weights is a linear feasibility problem, the discrete inverse-eigenvalue problem of a graph [14]. The sampler explores the harder global problem where the combinatorics also varies.

3.4 Obstruction: certification by free energy

The hard direction — proving ψ_U is *not* realizable — cannot be done by a finite search. In the sampling frame it becomes a statement about the measure, not the trajectory.

Theorem 3.5 (Gibbs concentration; Hwang). *Let $r^* = \inf_{K \in \mathcal{K}_L} r_U(K)$. As $\tau \downarrow 0$, $\pi_{\tau, \beta}$ concentrates weakly on the global minimizers of r_U over $\mathcal{K}_L$ [15]. Consequently, for any $\epsilon < r^*$, the realizing region $\{r_U \leq \epsilon\}$ is empty, and for ϵ slightly above r^* its posterior mass obeys*

$$\pi_\tau(\{r_U \leq \epsilon\}) \asymp e^{-I(\epsilon)/\tau}, \quad I(\epsilon) > 0 \text{ when } \epsilon < r^*, \quad (17)$$

with I a large-deviations rate (a free-energy gap).

Definition 3.6 (free-energy obstruction certificate). ψ_U is *obstructed* at level ϵ iff

$$\liminf_{\tau \rightarrow 0} \left[F_\tau(\{r_U \leq \epsilon\}) - F_\tau(\mathcal{K}_L) \right] > 0, \quad F_\tau(\cdot) = -\tau \log \pi_\tau(\cdot), \quad (18)$$

equivalently $r^* > \epsilon$: the residual provably floors above zero because the realizing region carries vanishing posterior mass.

This is the correct statistical-mechanics form of the “residual floor” certificate. It is *not* established by a finished search: simulated-annealing convergence [16, 17] is asymptotic and finite-time fallible. To certify obstruction one needs one of:

- (a) a quantitative **free-energy / large-deviations lower bound** $I(\epsilon) > 0$, estimated by controlled methods (nested sampling, annealed importance sampling, thermodynamic integration) *together with* a mixing-time guarantee;
- (b) a deterministic **sum-of-squares / Positivstellensatz** lower bound $r_U \geq r^* > 0$ via the Lasserre moment hierarchy [18] — the convex “foil” that gives a hard algebraic certificate where the residual is polynomial in the variables;
- (c) a **topological obstruction**: by Theorem 2.8, a harmonic target incompatible with the Betti numbers of every admissible bulk is unrealizable *a priori*, with no optimization at all. This is the cheapest certificate and should be checked first.

Remark 3.7 (unifying the two pictures). The convex lower bound (b) and the free-energy bound (a) are dual faces of one certificate: the moment–SOS hierarchy produces an algebraic identity witnessing $r \geq r^*$, while the free-energy route produces a partition-function ratio $\mathcal{Z}_{\{r \leq \epsilon\}}/\mathcal{Z} \rightarrow 0$. The former is hard but scales poorly; the latter is asymptotic but native to the sampler.

3.5 Summary of the theory

1. The fixed boundary is encoded in the *move set* (interior Pachner moves; Theorem 2.15), realizing the Dirichlet/Hartle–Hawking/cobordism template.
2. Realizability is witnessed by any low-residual sample (Proposition 3.2); it is generically non-unique.
3. Non-realizability is a free-energy gap / vanishing posterior mass (Theorem 3.5), certified by a large-deviations bound, an SOS bound, or — best — a topological incompatibility (Theorem 2.8).

4 Implementation plan

The construction is largely an assembly of existing components plus one structural addition (a fixed boundary in the interaction-history sampler). We list the workstreams by capability.

4.1 Existing components

Capability	Status	Locus
MI metric $\ell = -\log(I/I_{\max})$	built	interaction-history edge lengths
Regge action, deficit angles	built	<code>ReggeSolver</code> ; <code>cellHingeAction</code>
Metropolis acceptance $e^{-\beta\Delta S}$	built	interaction-history <code>accept</code>
Hodge Laplacian L_k , harmonics	built	<code>HodgeLaplacian</code>
Residual $r = \ (\mathcal{K} - P_\psi)L\psi\ ^2$	built	<code>EigenstateSynthesis</code>
Bending $\text{vec}(U)$, $\langle\psi_A U \psi_B\rangle$	built	<code>ChoiJamiolkowski</code>
Heat-kernel spectral dimension	built	spectral-dimension module
Boundary-fixed Pachner growth	built	<code>EigenstateSynthesis::growInterior</code>
Free-connectivity search ($k = 0$)	built (#201)	<code>RealizabilityOracle</code>
$k = 1$ triangle-candidate search	built (#203)	<code>RealizabilityOracle::decideHarmonic</code>
Fixed boundary in the sampler	missing	—
Residual term in MC acceptance	missing	—

4.2 Workstream: boundary pinning in the sampler

The interaction-history sampler currently has no immutable subcomplex (its growth frontier is consumed each step, and the reverse move may delete any cell). The realizability engine, by contrast, pins the boundary by an interior/boundary partition that the writers structurally respect. Port the latter pattern:

1. Seed an immutable boundary vertex/edge set once from the canonical boundary (codimension-one faces in exactly one top cell).
2. Exclude pinned vertices from the growth frontier, so forward moves cannot select them.
3. Guard the reverse move to abort if its deletion set meets the pinned subcomplex.
4. Carry ψ_U on the pinned boundary support (the lowest-index vertices, matching the embedding convention).

By Theorem 2.15 this preserves ergodicity over the admissible interiors.

4.3 Workstream: the residual term in acceptance

Extend the acceptance exponent (16) by $-\Delta r_U/\tau$: evaluate r_U via the existing residual routine, which reads live edge weights, so no new mesh plumbing is required; compute r_U^{after} in the throwaway-geometry pattern already used for ΔS so rejected proposals never mutate the live complex. Add an annealing schedule on τ .

4.4 Workstream: the move set

The forward/reverse interaction moves change the cell count (and therefore the scale at which d_S is read). Add fixed-volume interior *bistellar* moves so the chain can explore triangulations at fixed 4-volume, the regime in which Theorem 2.15 applies and in which the gravitational action is well-defined (fixed-volume sampling tames the conformal-mode unboundedness).

Choose the move set by degree. The free-connectivity result of Section 2.9 (internal #201,#203) fixes which moves can change the residual at all. If synthesis is performed at $k = 0$, a free interior-connectivity proposal (`attachInteriorVertex` with arbitrary incidences, scored by residual) strictly enlarges what is realizable and should be in the move set. If synthesis is at the Hodge-qubit degree $k = 1$ — the degree relevant to the cobordism/Dijkgraaf–Witten bridge — additive proposals are spectrally inert or boundary-locked, so the only effective boundary-fixed move is the stellar Pachner cone; the $k = 1$ search should therefore enumerate and log the (edge + triangle) candidates for transparency but expect them to be inert and fall back to the cone (“free = cone”). In either case the boundary stays bit-exact, so no proposal can perturb ψ_U on L .

4.5 Workstream: diagnostics and certificates

- *Realizability*: report the minimum r_U achieved and the witness geometry.
- *Obstruction*: estimate the posterior mass $\pi_\tau(\{r_U \leq \epsilon\})$ and the free-energy gap by thermodynamic integration / nested sampling; check the topological obstruction (Betti-number compatibility) first.
- *Geometry*: the spectral dimension running curve $d_S(\sigma)$, the Hausdorff dimension, and finite-size scaling — reported as diagnostics, with the evidentiary caveats of Section 5.

4.6 Experimental protocol

Validate on controls before any headline run: a *realizable* target (e.g. the constant boundary vector $\text{vec}(U) = (1, 1, 1, 1)$, an exact zero mode) must yield $r_U \rightarrow 0$ with the boundary fixed; an *obstructed* control must floor with a measurable free-energy gap. Only then study the geometry (spectral/Hausdorff dimension) of the realizing ensemble.

5 Honest assessment: novelty and the dimension bar

Novelty. The mutual-information metric (5) is due to [2]; Regge Monte Carlo and emergent spectral dimension are standard in dynamical triangulations [19]; emergent 4D networks under stochastic dynamics have been reported [21, 22]. The defensible contribution of the present construction is the *combination not previously assembled*: a *sampled* Regge ensemble over mutual-information-sourced edge lengths, with a *fixed boundary*, run in the *inverse* direction to synthesize a bulk realizing a prescribed operation, with the operation read off the Laplacian harmonic. The strongest framing

shows the present method *contains* [2] as its static, single-geometry, area-law limit. It should be positioned as a synthesis/methods advance, not as a discovery that mutual information defines a metric.

The dimension bar. An emergent-(3+1)D claim, by the standards of the quantum-gravity literature, requires (i) a *plateau* of $d_S(\sigma) \approx 4$ over a scale window, not a single crossing; (ii) the near-universal ultraviolet running toward $d_S \approx 2$ [19, 20]; (iii) a corroborating Hausdorff dimension $d_H \approx 4$ (branched polymers have $d_S = 4/3$ but $d_H = 2$, so d_S alone is insufficient); (iv) finite-size scaling showing the plateau widens with volume; and (v) identification of an extended, de Sitter-like phase rather than a crumpled or branched-polymer one. Emergent dimension should therefore enter this program as a careful secondary diagnostic, with these controls, rather than as the headline result.

References

- [1] M. Van Raamsdonk, *Building up spacetime with quantum entanglement*, Gen. Rel. Grav. **42**, 2323 (2010), arXiv:1005.3035.
- [2] C. Cao, S. M. Carroll, S. Michalakis, *Space from Hilbert Space: Recovering Geometry from Bulk Entanglement*, Phys. Rev. D **95**, 024031 (2017), arXiv:1606.08444.
- [3] (Leighton–Trudel), *Emergent Distance and Metricity of Mutual Information in 1D Quantum Chains*, arXiv:2507.09749 (2025).
- [4] T. Regge, *General relativity without coordinates*, Nuovo Cimento **19**, 558 (1961).
- [5] G. W. Gibbons, S. W. Hawking, *Action integrals and partition functions in quantum gravity*, Phys. Rev. D **15**, 2752 (1977). (Boundary term: York, Phys. Rev. Lett. **28**, 1082, 1972.)
- [6] G. W. Gibbons, S. W. Hawking, M. J. Perry, *Path integrals and the indefiniteness of the gravitational action*, Nucl. Phys. B **138**, 141 (1978).
- [7] J. B. Hartle, S. W. Hawking, *Wave function of the Universe*, Phys. Rev. D **28**, 2960 (1983).
- [8] M. F. Atiyah, *Topological quantum field theories*, Publ. Math. IHÉS **68**, 175 (1988).
- [9] U. Pachner, *P.L. homeomorphic manifolds are equivalent by elementary shellings*, Eur. J. Combin. **12**, 129 (1991).
- [10] M. R. Casali, *A note about bistellar operations on PL-manifolds with boundary*, Geom. Dedicata **56**, 257 (1995).
- [11] T. Kalelkar, A. Phanse, *Geometric bistellar moves relate geometric triangulations*, arXiv:1907.02643 (2019).
- [12] C. Gordon, D. Webb, S. Wolpert, *One cannot hear the shape of a drum*, Invent. Math. **110**, 1 (1992).
- [13] E. Blåsten, H. Isozaki, M. Lassas, J. Lu, *Gel’fand’s inverse problem for the graph Laplacian*, J. Spectral Theory (2021), arXiv:2101.10026.
- [14] S. Fallat, H. Gupta, J. C.-H. Lin, *Inverse eigenvalue problem for Laplacian matrices of a graph*, SIAM J. Matrix Anal. Appl. (2024), arXiv:2411.00292.
- [15] C.-R. Hwang, *Laplace’s method revisited: weak convergence of probability measures*, Ann. Probab. **8**, 1177 (1980).

- [16] S. Geman, D. Geman, *Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images*, IEEE TPAMI **6**, 721 (1984).
- [17] B. Hajek, *Cooling schedules for optimal annealing*, Math. Oper. Res. **13**, 311 (1988).
- [18] J. B. Lasserre, *Global optimization with polynomials and the problem of moments*, SIAM J. Optim. **11**, 796 (2001).
- [19] J. Ambjørn, J. Jurkiewicz, R. Loll, *The spectral dimension of the Universe is scale dependent*, Phys. Rev. Lett. **95**, 171301 (2005), arXiv:hep-th/0505113.
- [20] S. Carlip, *Dimension and dimensional reduction in quantum gravity*, Class. Quantum Grav. **34**, 193001 (2017), arXiv:1705.05417.
- [21] P. Tee, *Dynamics and the emergence of geometry in an information mesh*, Eur. Phys. J. C **80** (2020), arXiv:1909.05811.
- [22] C. A. Trugenberger, *Quantum gravity as an information network: self-organization of a 4D universe*, Phys. Rev. D **92**, 084014 (2015), arXiv:1501.01408.